

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

**Product Identifier**

Perihalan Produk: **Copper single crystal**  
Product Description: **Copper single crystal**  
Cat No. : 13295  
CAS No 7440-50-8  
Molecular Formula Cu

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
Selangor Darul Ehsan, Malaysia.  
Main line: +60 3-5525 7888

**Supplier**

**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number** Tel: +03-5525 7888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**

**Classification of the substance or mixture**

**Label Elements**

**Hazard Statements**

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other Hazards**

Toxic to terrestrial vertebrates

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

Toxicity to Soil Dwelling Organisms

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Copper    | 7440-50-8 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Copper oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV                  | OSHA PEL                                                                                       |
|-----------|----------|----------------------------|------------------------------------------------------------------------------------------------|
| Copper    |          | TWA: 0.2 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> |

| Component | European Union | The United Kingdom                                                                                                                         | Germany                                                                           |
|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Copper    |                | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr | TWA: 0.01 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.02 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Respiratory Protection**  
**Recommended Filter type:**

No protective equipment is needed under normal use conditions  
Particle filter

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls**

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

**Information on basic physical and chemical properties**

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| Appearance                                     | Brown                    |                                          |
| Physical State                                 | Solid                    |                                          |
| Odor                                           | Odorless                 |                                          |
| Odor Threshold                                 | No data available        |                                          |
| pH                                             | Not applicable           |                                          |
| Melting Point/Range                            | 1083 °C / 1981.4 °F      |                                          |
| Softening Point                                | No data available        |                                          |
| Boiling Point/Range                            | 2595 °C / 4703 °F        | @ 760 mmHg                               |
| Flash Point                                    | No information available | <b>Method -</b> No information available |
| Evaporation Rate                               | Not applicable           | Solid                                    |
| Flammability (solid,gas)                       | No information available |                                          |
| Explosion Limits                               | No data available        |                                          |
| Vapor Pressure                                 | No data available        |                                          |
| Vapor Density                                  | Not applicable           | Solid                                    |
| Specific Gravity / Density                     | No data available        |                                          |
| Bulk Density                                   | No data available        |                                          |
| Water Solubility                               | Insoluble                |                                          |
| Solubility in other solvents                   | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| Autoignition Temperature                       | No data available        |                                          |
| Decomposition Temperature                      | No data available        |                                          |
| Viscosity                                      | Not applicable           | Solid                                    |
| Explosive Properties                           | No information available |                                          |
| Oxidizing Properties                           | No information available |                                          |

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

Molecular Formula Cu  
Molecular Weight 63.54

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Air sensitive.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation. Exposure to air.

### Incompatible Materials

Strong oxidizing agents. Acids.

### Hazardous Decomposition Products

Copper oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

**Product Information** No acute toxicity information is available for this product

(a) acute toxicity;  
Oral No data available  
Dermal No data available  
Inhalation No data available

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation              |
|-----------|-----------|-------------|------------------------------|
| Copper    | -         | -           | LC50 > 5.11 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory</b>                                | No data available                                                                                                                   |
| <b>Skin</b>                                       | No data available                                                                                                                   |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                   |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product                                                      |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                   |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                                                                                   |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                   |
| <b>Target Organs</b>                              | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | Not applicable<br>Solid                                                                                                             |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | No information available.                                                                                                           |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

|                                   |                                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Ecotoxicity effects</u></b> | Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system. |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                  | Water Flea                                    | Freshwater Algae                                                                                                                                   | Microtox |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Copper    | LC50: = 1.25 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 0.8 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.112 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: = 0.052 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.0068 - 0.0156 mg/L, 96h (Pimephales promelas)<br>LC50: < 0.3 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.2 mg/L, 96h | EC50: = 0.03 mg/L, 48h Static (Daphnia magna) | EC50: 0.031 - 0.054 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 0.0426 - 0.0535 mg/L, 72h static (Pseudokirchneriella subcapitata) |          |

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

|  |                                       |  |  |  |
|--|---------------------------------------|--|--|--|
|  | flow-through<br>(Pimephales promelas) |  |  |  |
|--|---------------------------------------|--|--|--|

|                                              |                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Persistence and degradability</u></b>  | Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary<br>Insoluble in water, May persist.<br>Not relevant for inorganic substances.<br>Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |
| <b>Persistence</b>                           |                                                                                                                                                                                                                                                                                                                    |
| <b>Degradability</b>                         |                                                                                                                                                                                                                                                                                                                    |
| <b>Degradation in sewage treatment plant</b> |                                                                                                                                                                                                                                                                                                                    |

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b><u>Bioaccumulative potential</u></b> | May have some potential to bioaccumulate; Product has a high potential to bioconcentrate |
|-----------------------------------------|------------------------------------------------------------------------------------------|

|                                |                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| <b><u>Mobility in soil</u></b> | Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility. |
|--------------------------------|------------------------------------------------------------------------------------------------------------|

|                                               |                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------|
| <b><u>Endocrine Disruptor Information</u></b> | This product does not contain any known or suspected endocrine disruptors |
|-----------------------------------------------|---------------------------------------------------------------------------|

|                                     |                          |
|-------------------------------------|--------------------------|
| <b><u>Other adverse effects</u></b> | No information available |
|-------------------------------------|--------------------------|

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b><u>Waste treatment methods</u></b>      |                                                                                               |
| <b>Waste from Residues/Unused Products</b> | Dispose of in accordance with local regulations                                               |
| <b>Contaminated Packaging</b>              | Empty containers should be taken to an approved waste handling site for recycling or disposal |
| <b>Other Information</b>                   | Do not flush to sewer                                                                         |

## SECTION 14: TRANSPORT INFORMATION

|                                       |                                 |
|---------------------------------------|---------------------------------|
| <b><u>IMDG/IMO</u></b>                | Not regulated                   |
| <b><u>Road and Rail Transport</u></b> | Not regulated                   |
| <b><u>IATA</u></b>                    | Not regulated                   |
| <b>Special Precautions for User</b>   | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### **Safety, health and environmental regulations/legislation specific for the substance or mixture**

|                                  |            |
|----------------------------------|------------|
| <b>International Inventories</b> | X = listed |
|----------------------------------|------------|

| Component | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|-----------|--------|------|-----|-------|------|------|-------|------|------|
|-----------|--------|------|-----|-------|------|------|-------|------|------|

# SAFETY DATA SHEET

Copper single crystal

Revision Date 24-Mar-2025

|        |           |   |   |   |   |  |   |   |          |
|--------|-----------|---|---|---|---|--|---|---|----------|
| Copper | 231-159-6 | X | X | X | X |  | X | X | KE-08896 |
|--------|-----------|---|---|---|---|--|---|---|----------|

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

24-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**